

APPLICATION-2-BROADCAST-BRIDGE

PROVISIONAL PATENT APPLICATION

BROADCAST-TO-UNICAST MEDIA BRIDGE WITH FEC PRESERVATION AND HIERARCHICAL FEC CATALOG SIGNALING

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FIELD OF THE INVENTION

The present invention relates to media delivery systems that bridge broadcast television infrastructure to unicast streaming protocols, and to methods for signaling forward error correction parameters in media streaming catalogs with hierarchical configuration.

BACKGROUND OF THE INVENTION

Broadcast Television and MMTP

ATSC 3.0 (Advanced Television Systems Committee) is the next-generation broadcast television standard used in the Americas and South Korea. ARIB STD-B60 is the equivalent standard used in Japan. Both standards use MPEG Media Transport Protocol (MMTP, ISO 23008-1) for media delivery over broadcast networks.

Broadcast media is transmitted via Source-Specific Multicast (SSM) using MMTP packets that carry application-layer FEC metadata, including Source FEC Payload IDs for source symbols and Repair FEC Payload IDs for repair symbols. FEC parameters are signaled via the S-TSID (Service-based Transport Session Instance Description), an XML document containing FECParameters elements with RaptorQ OTI (Object Transmission Information), symbol sizes, and block configurations.

Media over QUIC (MoQ)

Media over QUIC (MoQ) is an emerging IETF protocol for real-time media delivery using a publish/subscribe model over QUIC and WebTransport. MoQ uses JSON-based catalogs to describe available media tracks, codecs, and configurations. Subscribers fetch catalogs to discover available content before establishing subscriptions.

FEC Signaling in Prior Art

Existing FEC signaling methods include:

- **SDP (RFC 6364):** Session Description Protocol attributes for FEC Framework configuration. Requires SIP or RTSP session establishment. Per-flow configuration without hierarchical defaults.
- **ATSC S-TSID (A/331):** XML-based signaling with per-RepairFlow FEC parameters. Not hierarchical. Not accessible via HTTP.
- **DASH MPD:** Has hierarchical structure (Period, AdaptationSet, Representation) with property inheritance, but defines NO FEC-specific properties.
- **MoQ Catalog (I-D.ietf-moq-catalogformat):** JSON-based catalog for MoQ track discovery. Contains ZERO provisions for FEC parameter signaling.

Limitations of Prior Art

1. **No broadcast-to-MoQ bridge exists.** ATSC 3.0 gateways (e.g., ADTH, HDHomeRun) convert broadcast to local network streaming but do not output MoQ tracks, do not preserve FEC parameters, and do not convert S-TSID to MoQ catalog format.
 2. **No S-TSID to MoQ catalog conversion.** S-TSID (XML) and MoQ catalog (JSON) are structurally different. No standard or known system defines a bidirectional conversion between them.
 3. **No hierarchical FEC catalog.** FEC parameters in streaming manifests are either session-level (SDP), per-flow (S-TSID), or absent (DASH, MoQ). No system provides top-level FEC defaults with per-track overrides in a hierarchically structured catalog.
 4. **No pre-subscription FEC discovery.** In SDP-based systems, FEC parameters are exchanged during session negotiation. Subscribers cannot discover FEC availability before committing to a session.
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SUMMARY OF THE INVENTION

The present invention provides two interrelated systems:

1. An end-to-end broadcast-to-unicast bridge that receives broadcast media via SSM multicast, performs FEC recovery, converts broadcast signaling (S-TSID) to streaming catalog (MoQ JSON), and re-publishes media as MoQ tracks — enabling subscribers without broadcast receiver hardware, MMTP decoders, or FEC processing capability to consume broadcast content.
 2. A hierarchical FEC catalog signaling method where FEC parameters are specified at top-level (defaults) and per-track (overrides) in a JSON catalog, discoverable via HTTP before subscription, with in-band control message override and support for per-track algorithm selection including a “none” mode with jitter buffering.
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DETAILED DESCRIPTION OF THE INVENTION

First Embodiment: Broadcast-to-Unicast Bridge

FIG. 1 illustrates the broadcast-to-unicast bridge architecture.

Component 101 - Broadcast Transmitter: An ATSC 3.0 or ARIB STD-B60 broadcast transmitter emitting MMTP packets via SSM multicast. The transmitter generates AL-FEC repair symbols using RaptorQ (RFC 6330) and transmits both source and repair MMTP packets on the multicast group.

Component 102 - S-TSID Signaling: The broadcast system distributes S-TSID signaling containing: - Logical Stream (LS) elements describing media tracks - FECParameters elements with: fecSchemeID, symbolSize, maxSourceBlockLength (K), maxNumberEncSymbols (K+P) - RepairFlow elements mapping repair data to source streams - RS (Receiving Station) elements with multicast group addresses

Component 103 - Bridge Gateway: The bridge gateway performs:

Step 103a - Multicast Reception: The gateway joins the SSM multicast group and receives MMTP packets (both source and repair).

Step 103b - FEC Recovery: The gateway performs RaptorQ or Reed-Solomon FEC decoding on received blocks. For blocks where all K source symbols arrived cleanly, no decoding is necessary. For blocks with missing source symbols, the gateway uses available repair symbols to recover the missing data.

Step 103c - S-TSID to Catalog Conversion: The gateway converts broadcast signaling to a MoQ JSON catalog using the following field mapping:

S-TSID Field	MoQ Catalog Field
LS (Logical Stream)	tracks[] entry
LS.tsi (Transport Session ID)	tracks[].name
FECParameters.fecSchemeID	tracks[].fec.algorithm (scheme 6 -> "raptorq")
FECParameters.symbolSize	tracks[].fec.symbolSize
FECParameters.maxSourceBlockLength	tracks[].fec.sourceSymbols
FECParameters.maxNumberEncSymbols - maxSourceBlockLength	tracks[].fec.repairSymbols

S-TSID Field	MoQ Catalog Field
RepairFlow.tsi	<code>tracks[].fec.repairTrack</code> (gateway assigns track name)
FECParameters.maximumDelay	<code>tracks[].fec.interleaveDepth</code> (converted to ms)
RS.source_address	<code>multicast.endpoints[].sourceAddress</code>
RS.group_address	<code>multicast.endpoints[].groupAddress</code>
RS.port	<code>multicast.endpoints[].port</code>

The reverse conversion (MoQ catalog to S-TSID) uses the inverse mapping, enabling MoQ-originated content to be retransmitted via broadcast infrastructure.

Step 103d - MoQ Publication: The gateway publishes FEC-recovered source media as MoQ tracks over QUIC/WebTransport. Each MMTP source track becomes a MoQ track. The gateway may optionally: - Re-publish both source and repair tracks (for subscribers wanting FEC protection on the unicast path) - Re-package MMTP media into CMAF (fMP4) segments for compatibility with HLS/DASH clients - Strip MMTP headers and deliver raw codec bitstream

Component 104 - MoQ Subscribers: Subscribers receive media via standard MoQ subscription without requiring MMTP decoders, FEC processing, or broadcast receiver hardware. They interact with the bridge gateway as they would with any MoQ publisher.

Step 103e - Bidirectional Operation: The reverse path enables MoQ publishers to reach broadcast infrastructure: 1. MoQ subscriber sends media to the bridge gateway 2. Gateway converts MoQ catalog to S-TSID 3. Gateway packages media as MMTP packets with AL-FEC 4. Gateway transmits via SSM multicast

Second Embodiment: Hierarchical FEC Catalog Signaling

FIG. 2 illustrates the hierarchical FEC catalog structure and precedence model.

Level 1 - Top-Level Defaults (201): The catalog JSON document contains a top-level `fec` object specifying default FEC parameters:

```
{
  "fec": {
    "algorithm": "raptorq",
    "sourceSymbols": 32,
    "repairSymbols": 8,
    "symbolSize": 1312,
    "interleaveDepth": 133
  },
  "tracks": [ ... ]
}
```

These defaults apply to all tracks that do not specify their own FEC configuration.

Level 2 - Per-Track Overrides (202): Individual tracks may override specific fields from the top-level defaults:

```
{
  "tracks": [
    {
      "name": "video",
      "fec": {
        "repairTrack": "video/repair"
      }
    },
    {
      "name": "audio",
      "fec": {
        "algorithm": "none",
        "jitterBufferMs": 50
      }
    }
  ]
}
```

The video track inherits all top-level FEC parameters and adds its repair track name. The audio track overrides the algorithm to “none” with a jitter buffer, indicating no FEC encoding is applied and the receiver should use a 50ms reorder buffer.

Level 3 - In-Band Override (203): After subscription establishment, an in-band FEC_CONFIG control message may be sent from publisher to subscriber. This message carries authoritative FEC parameters that override both per-track and top-level catalog values.

Precedence Order: FEC_CONFIG message > per-track catalog > top-level catalog defaults

Step 204 - HTTP Pre-Discovery: The catalog is fetchable via HTTP (e.g., GET /catalog.json) before establishing a MoQ session. This enables subscribers to: - Discover whether FEC is available for a track - Determine the FEC algorithm and parameters - Decide whether to subscribe to repair tracks - Estimate bandwidth overhead (repairSymbols / sourceSymbols)

All before committing to a QUIC connection or subscription.

Step 205 - “None” Algorithm: The algorithm value “none” (registry value 0x00) indicates no FEC encoding. When present, source data is delivered without repair symbols. The optional jitterBufferMs field specifies a receiver-side reorder buffer in milliseconds, providing tolerance for out-of-order packet delivery without FEC overhead.

Use cases for “none”: - Audio tracks where codec-level Packet Loss Concealment (PLC) adequately handles gaps - Ultra-low-latency (ULL) applications where any FEC latency is unacceptable (e.g., financial data feeds) - Reliable QUIC stream delivery where retransmission is sufficient

Step 206 - Per-Track Algorithm Selection: Different media types within the same session can use different FEC algorithms:

Media Type	Recommended Algorithm	Rationale
Video	raptorq	Large keyframes expensive to lose
Audio	none	Small frames, PLC handles gaps
Subtitles	none	Reliable QUIC sufficient

This minimizes total FEC overhead by applying protection only where recovery provides significant benefit.

Step 207 - Deferred Configuration Merging: In implementation, FEC parameters may be set before track renditions exist in the catalog (e.g., when the encoder signals FEC configuration before the first media frame). The system maintains a pending FEC configuration that is merged incrementally: 1. set_repair_track("video/repair") sets repair_track in pending 2. set_fec_config("raptorq", 32, 8, 1312, 133) merges algorithm, K, P, T, depth into

pending WITHOUT overwriting repair_track 3. When a track rendition is created, the pending configuration is cloned to the rendition

This ordering-independent merge ensures that parameters set by different system components at different times are correctly combined.

CLAIMS

Independent Claim 1

A system for bridging broadcast media to unicast streaming, comprising:

- a. a broadcast gateway receiving media packets with application-layer forward error correction via source-specific multicast from a broadcast transmitter;
- b. said gateway performing FEC decoding on received packets to recover media lost during broadcast transmission;
- c. said gateway converting broadcast signaling from a first format to a streaming catalog in a second format, wherein said conversion maps:
 - stream elements to catalog track entries;
 - FEC parameters to catalog FEC extension fields, including mapping a scheme identifier to an algorithm string, a symbol size to a symbolSize integer, a source block length to a sourceSymbols integer, and a difference between total encoding symbols and source block length to a repairSymbols integer;
 - repair flow elements to catalog repair track entries;
- d. said gateway publishing recovered source media as tracks in a publish/subscribe session over a reliable transport protocol;
- e. one or more subscribers receiving said tracks without requiring capability to decode said media packets in their original format, without requiring FEC processing, and without requiring broadcast receiver hardware.

Dependent Claim 2

The system of Claim 1, wherein said first format is S-TSID XML conforming to ATSC A/331 and said second format is JSON conforming to a MoQ catalog format.

Dependent Claim 3

The system of Claim 1, further comprising the reverse conversion from said second format to said first format, enabling content originated in the publish/subscribe system to be retransmitted via broadcast infrastructure.

Dependent Claim 4

The system of Claim 1, wherein said gateway re-packages recovered media from a first container format to a second container format, producing segments compatible with HTTP adaptive streaming protocols.

Dependent Claim 5

The system of Claim 1, wherein said gateway selectively publishes both source and repair tracks for subscribers that request FEC protection, while publishing source-only tracks for subscribers that do not.

Dependent Claim 6

The system of Claim 1, wherein said gateway is deployed at an ISP edge router with inline multicast tunneling capability, forming a hierarchical distribution node that receives broadcast via tunnel and re-publishes to local subscribers.

Independent Claim 7

A method for forward error correction parameter signaling in a media streaming system, comprising:

- a. providing a catalog document containing:
 - a top-level FEC configuration object specifying default parameters including an algorithm identifier, a source symbol count, a repair symbol count, a symbol size, and an interleave depth;

- one or more per-track FEC configuration objects, each capable of overriding specific fields from said top-level defaults, wherein fields not specified in a per-track object inherit values from said top-level object;
- b. making said catalog document available via HTTP prior to establishment of a streaming session, enabling subscribers to discover FEC parameters before subscription;
- c. after subscription establishment, optionally sending an in-band control message carrying authoritative FEC parameters;
- d. applying a precedence order wherein said in-band control message overrides per-track catalog values, which override top-level catalog defaults.

Dependent Claim 8

The method of Claim 7, wherein said algorithm identifier includes a “none” value indicating no FEC encoding, and said per-track FEC configuration object for a track with algorithm “none” includes a jitterBufferMs field specifying a receiver-side reorder buffer in milliseconds.

Dependent Claim 9

The method of Claim 7, wherein different tracks within the same catalog specify different algorithm identifiers, enabling per-media-type FEC selection within a single streaming session.

Dependent Claim 10

The method of Claim 7, further comprising maintaining a deferred FEC configuration that is merged incrementally as track renditions are created, wherein parameters set by separate configuration operations at different times are combined without overwriting previously set fields.

Dependent Claim 11

The method of Claim 7, wherein said interleave depth is expressed in milliseconds, decoupling the FEC block span from media frame rate and enabling the same catalog value to apply to media types with different frame rates.

Dependent Claim 12

The method of Claim 7, wherein for multicast delivery where no back-channel exists for said in-band control message, said catalog FEC parameters serve as the sole authoritative configuration source.

ABSTRACT

A system and method for bridging broadcast television (ATSC 3.0, ARIB STD-B60) to unicast streaming (MoQ/WebTransport) with forward error correction preservation. A gateway receives MMTP broadcast media via SSM multicast, performs FEC recovery, converts broadcast signaling (S-TSID XML) to a streaming catalog (MoQ JSON) with algorithmic field mapping, and re-publishes recovered media as publish/subscribe tracks. Subscribers consume content without requiring broadcast hardware, MMTP decoders, or FEC capability. A hierarchical FEC catalog signaling method provides top-level defaults with per-track overrides in a JSON catalog, HTTP-accessible before subscription, with in-band control message override. Per-track algorithm selection supports mixed FEC schemes (e.g., video=raptorq, audio=none) and a “none” mode with receiver-side jitter buffering.

FIGURES

FIG. 1: Broadcast-to-unicast bridge architecture showing broadcast transmitter, SSM multicast path, bridge gateway (with FEC recovery, S-TSID conversion, MoQ publication), and MoQ subscribers.

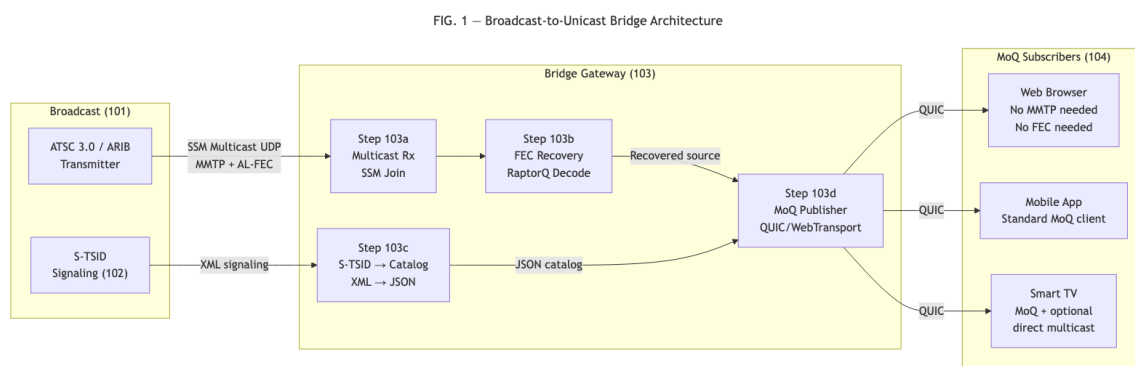


FIG. 1 — Broadcast-to-Unicast Bridge

FIG. 2: Hierarchical FEC catalog structure showing three precedence levels (top-level defaults, per-track overrides, in-band FEC_CONFIG message) with inheritance arrows and override semantics.

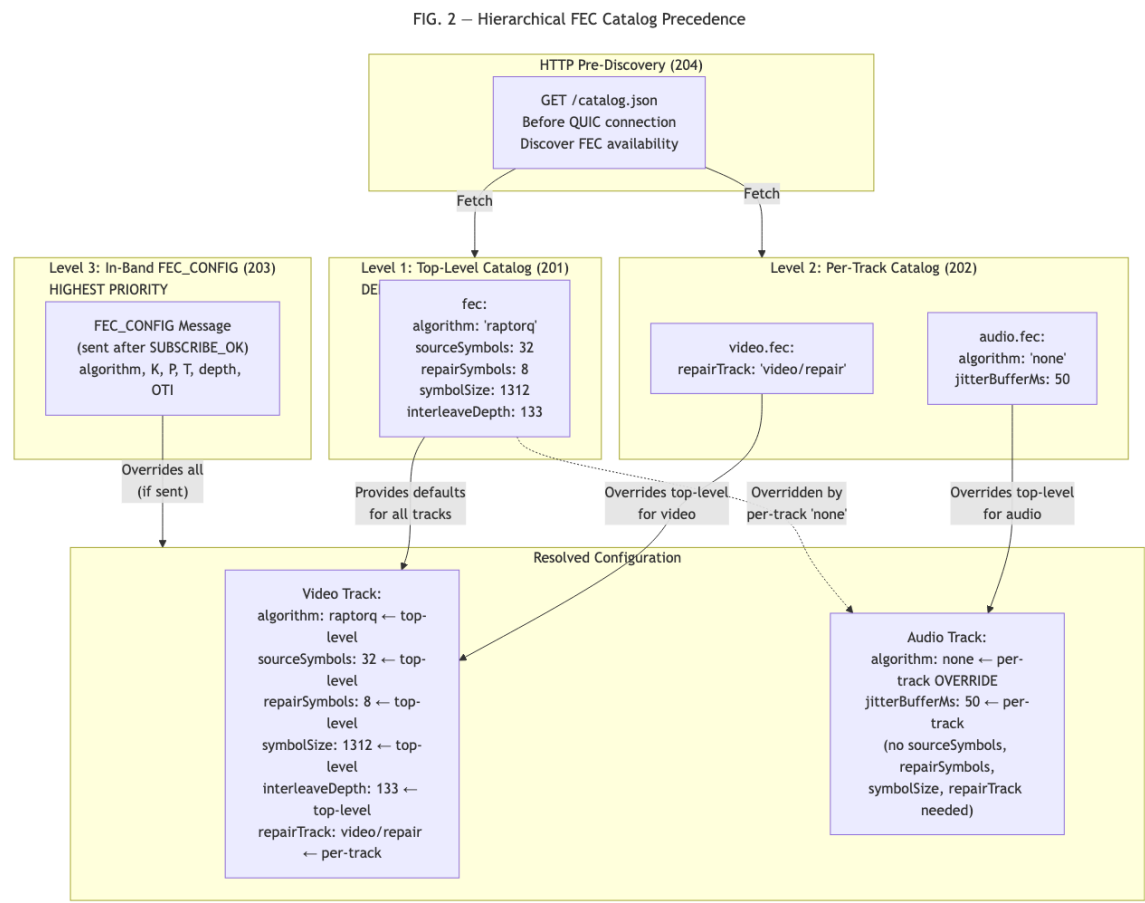


FIG. 2 — Hierarchical FEC Catalog

FIG. 3: S-TSID to MoQ catalog conversion field mapping diagram showing XML elements mapped to JSON fields.

FIG. 3 — S-TSID to MoQ Catalog Field Mapping

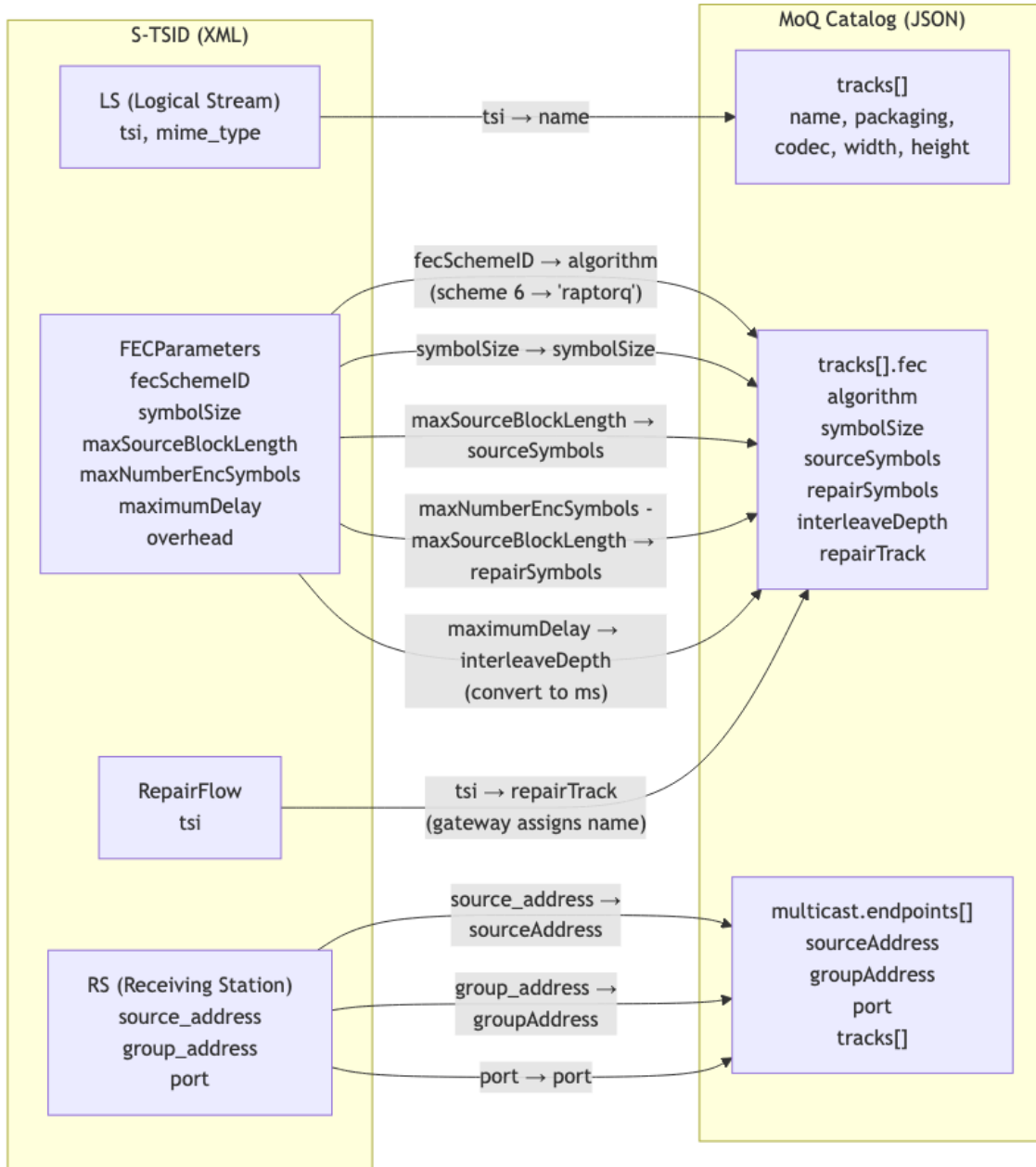


FIG. 3 — S-TSID to Catalog Conversion

FIG. 4: Deferred FEC configuration merge sequence showing three configuration calls and their cumulative effect on pending and rendition FEC state.

FIG. 4 – Deferred FEC Configuration Merge Sequence

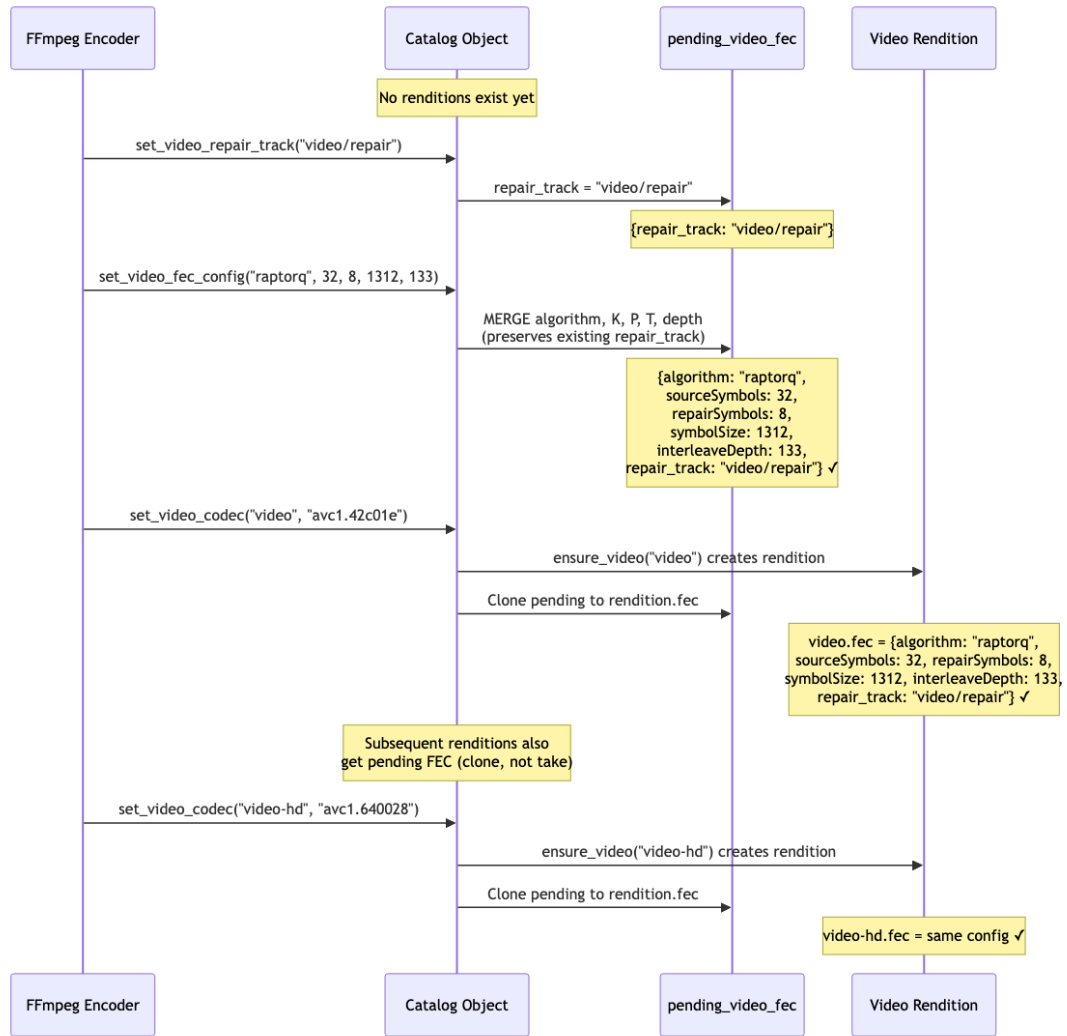


FIG. 4 — Deferred Configuration Merge